Solartron 7075 opto-isolated USB interface

USB-B -> RP2354A -> 6 opto channels -> shift registers -> 50-way D (SKB of the 70754 BCD interface unit)

MCU_USB

File: mcu_usb.kicad_sch

ISOLATION

File: isolation.kicad_sch

DVM_INPUTS

File: dvm_inputs.kicad_sch

DVM_OUTPUTS

File: dvm_outputs.kicad_sch

Sheets are connected with global labels.

Signals crossing the isolation barrier (3 x HCPL-2631 dual 10 MBd optocouplers):

SCK	MCU -> DVM side	shift clock for both register chains (rising edge shifts)
MOSI	MCU -> DVM side	command bits into 2 x TPIC6C595 (16 open-drain outputs)
LATCH	MCU -> DVM side	low = load 5 x 74HCT165 (40 inputs); rising edge = update command outputs
OEn	MCU -> DVM side	low = enable command outputs (high / LED off = all released)
MISO	DVM side -> MCU	status / BCD data out of the 74HCT165 chain
DRDY	DVM side -> MCU	PRINT command level (SKB pin 34): high = reading complete, data valid

Power: USB VBUS -> AP2112K-3.3 (MCU side). VBUS -> ferrite -> isolated 1 W DC-DC (9 V) -> L78M05 -> +5V_ISO (the 50-way connector has no supply pin, so the isolated logic is powered across the barrier).

Mechanical: 116 x 50 mm, 4 layers. J1 (DD-50 plug) is on the bottom side, everything else on the top side; the board is held by two 4-40 UNC screws through J1's flange holes into the jackposts of SKB.

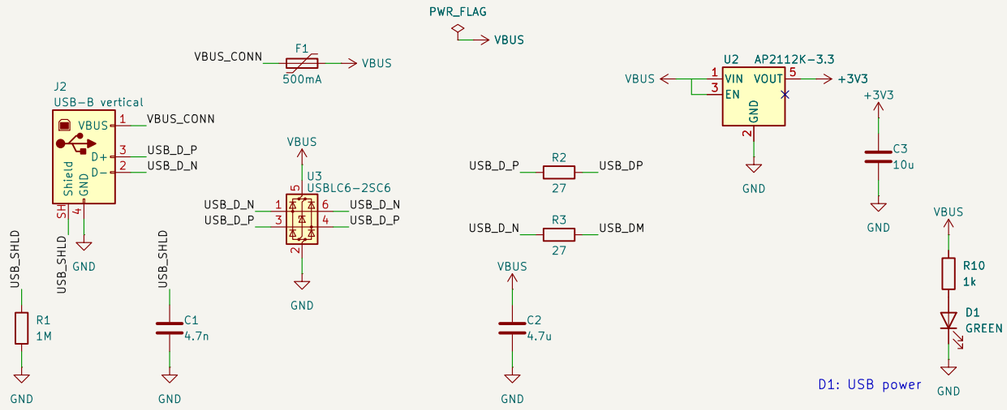
Plugs into SKB (50-way) of the 70754 Parallel BCD Interface Unit
Generated by scripts/gen_sch.py
Solartron 7075 USB interface (RP2354A)

Sheet: /
File: solartron7075_usb_interface.kicad_sch

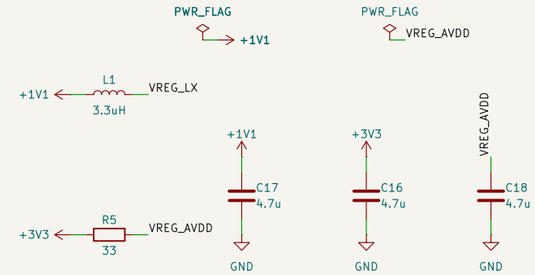
Title: Solartron 7075 opto-isolated USB interface

Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 1/5

USB-B input, ESD protection, 3.3 V LDO (USB / PC ground domain)

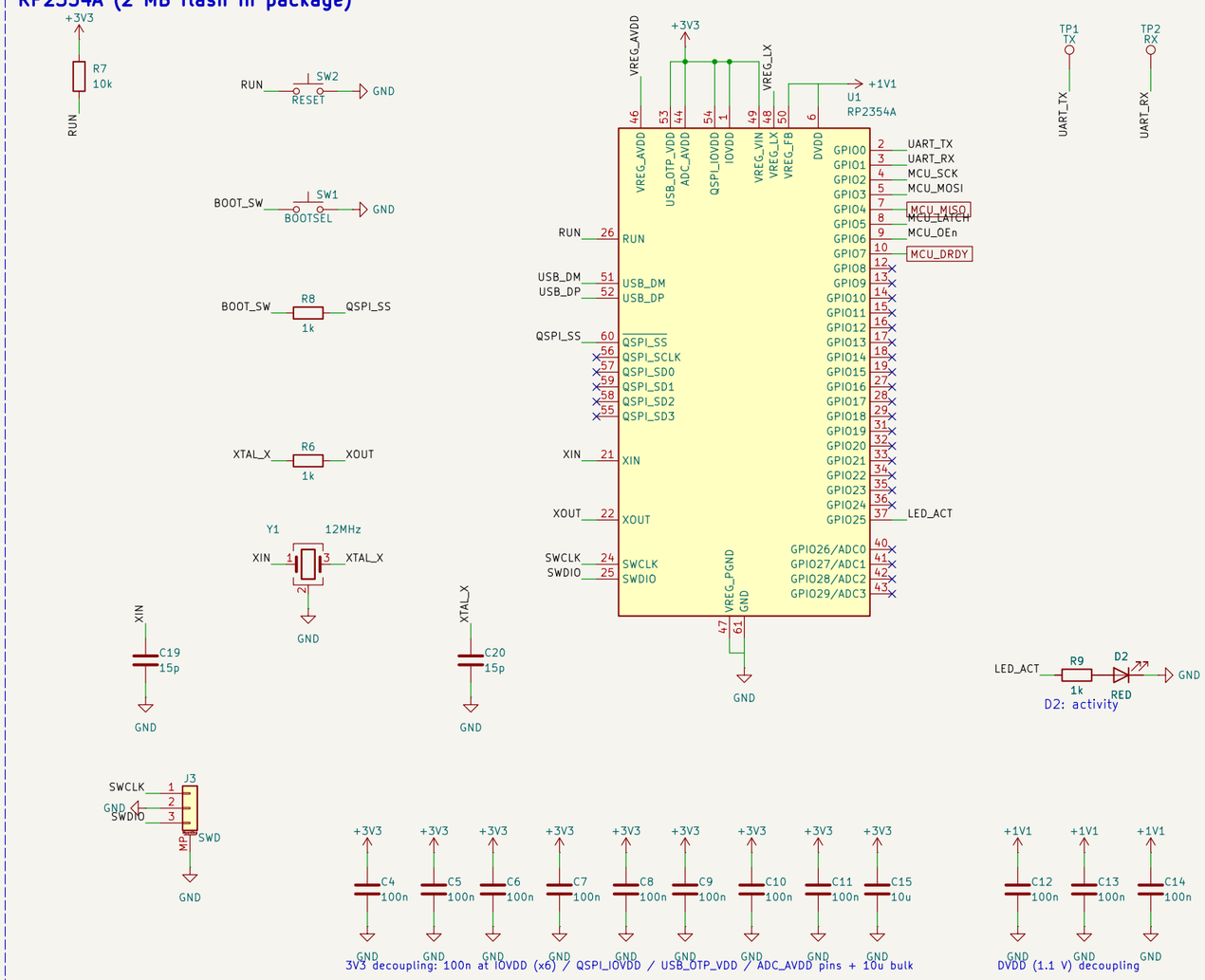


Core 1.1 V switching regulator (RP2350 reference circuit)

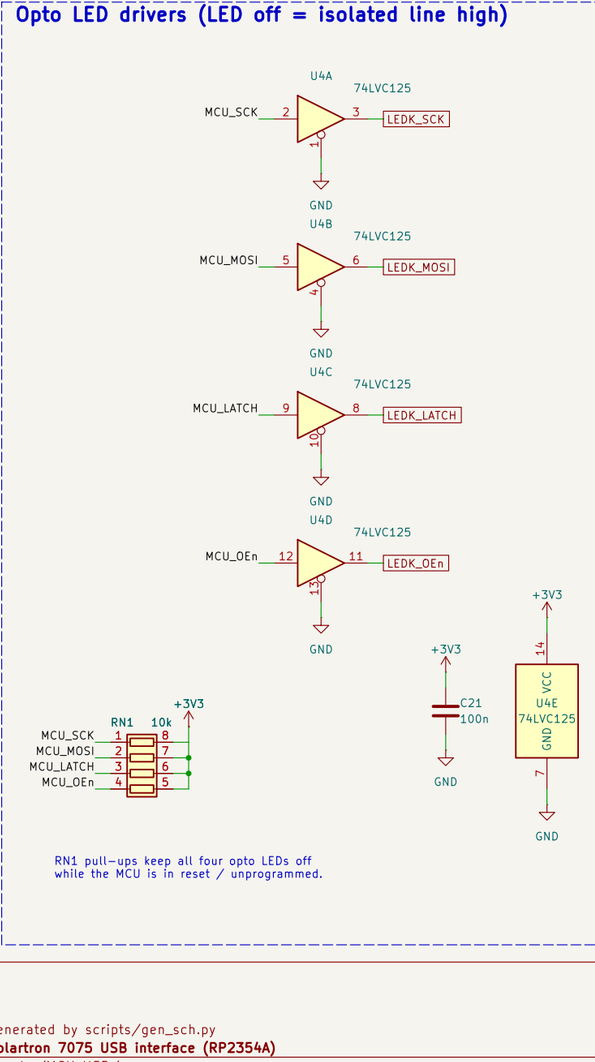


C16 at VREG_VIN, C17 at L1 output, C18 + R5 filter VREG_AVDD.
L1 is polarised: follow the RP2350 datasheet layout.

RP2354A (2 MB flash in package)



Opto LED drivers (LED off = isolated line high)



RN1 pull-ups keep all four opto LEDs off
while the MCU is in reset / unprogrammed.

Generated by scripts/gen_sch.py

Solartron 7075 USB interface (RP2354A)

Sheet: /MCU_USB/
File: mcu_usb.kicad_sch

Title: RP2354A microcontroller, USB and 3.3 V supply

Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 2/5

USB / MCU ground domain (GND, +3V3, VBUS)

Isolated multimeter domain (GND_ISO, +5V_ISO)

ISOLATION BARRIER

U5 HCPL-2631

U6 HCPL-2631

U7 HCPL-2631

180R: IF = 9 mA typ from 3.3 V

MCU -> multimeter side: SPI-style clock/data, register latch, output enable.
LED off (MCU in reset, USB unplugged) = line high on the isolated side.

Multimeter -> MCU: serial status data and PRINT (data ready) level.
U7 output stage runs from VBUS (5 V); open collectors pull up to 3.3 V.

Uplink LED drivers (TTL-level inputs)

390R: IF = 8.5 mA typ from +5V_ISO

Isolated supply: USB 5 V -> 9 V isolated DC-DC -> 5 V linear regulator

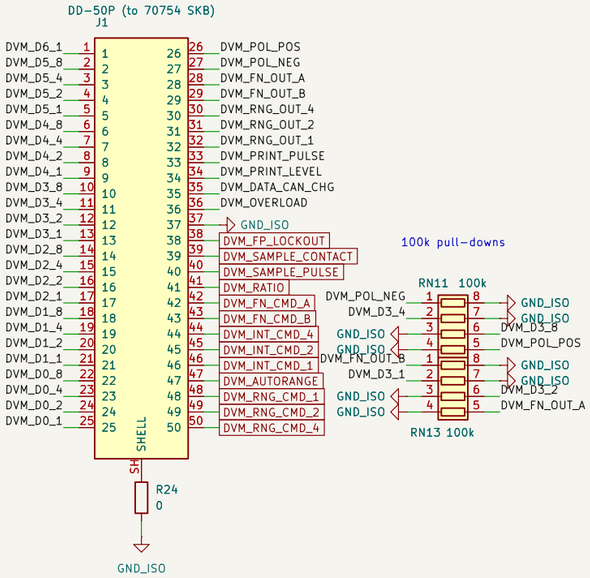
D3: isolated supply OK

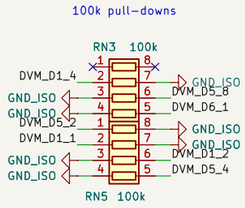
Generated by scripts/gen_sch.py
Solartron 7075 USB interface (RP2354A)
Sheet: /ISOLATION/
File: isolation.kicad_sch

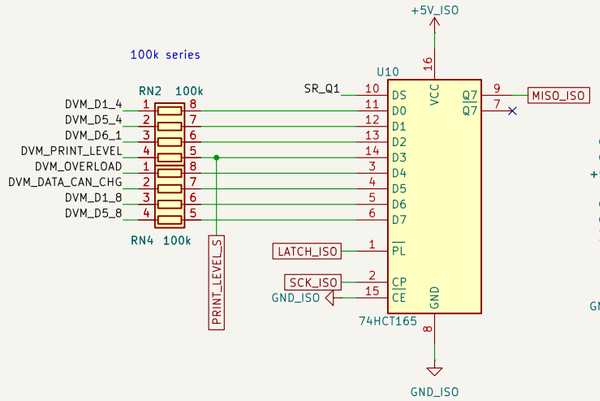
Title: Opto-isolation barrier and isolated 5 V supply

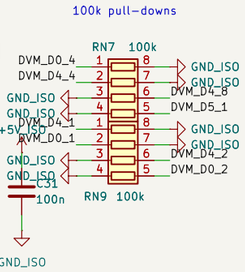
Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 3/5

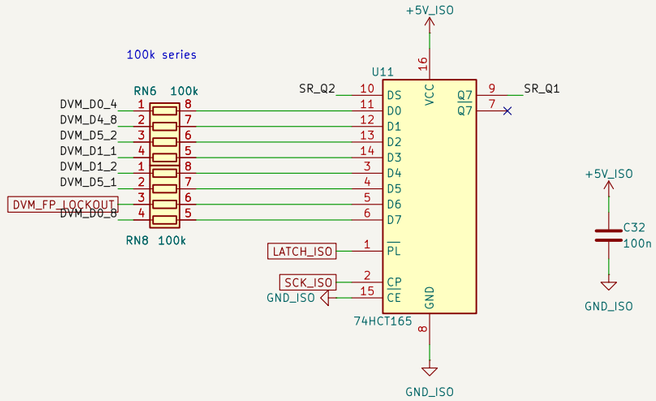
J1 mounts on the BOTTOM side and plugs straight into
the "Systems Interface" socket SKB of the 70754 unit.
Pin numbers = SKB pin numbers (manual page 9.9).

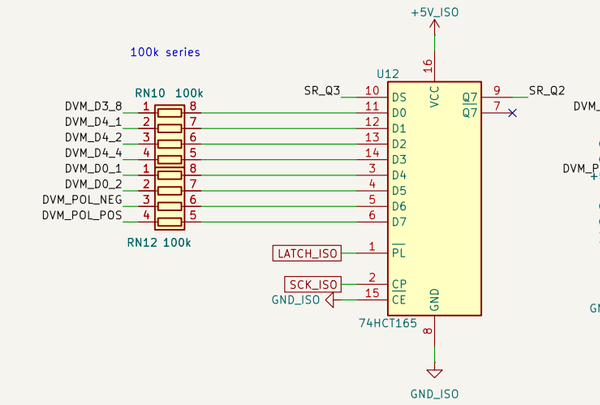


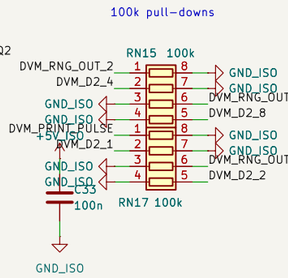


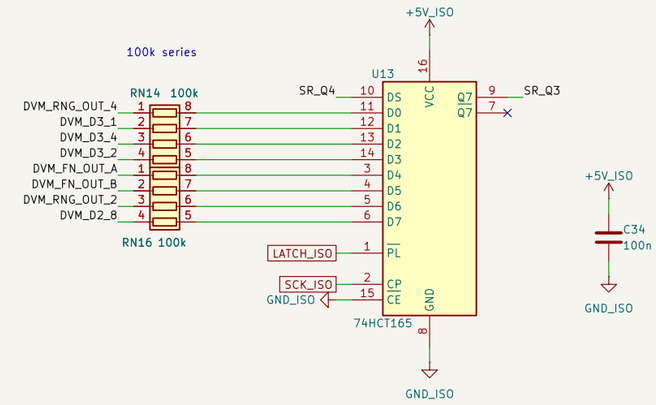




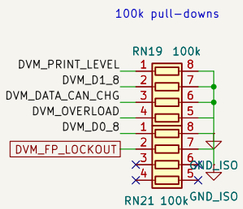


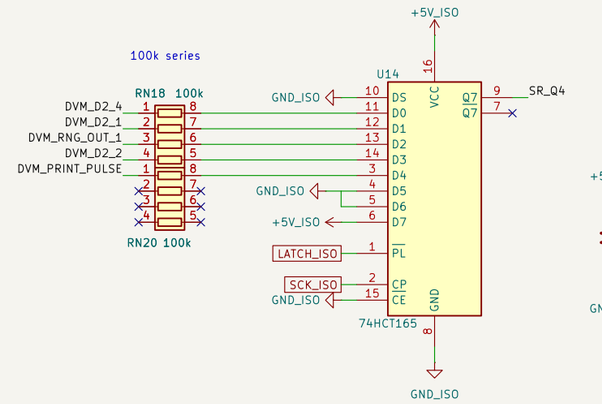






R24: connector shell / jackscrews to GND_ISO.
SKB pin 37 = multimeter EARTH / logic 0.
Remove R24 to float the shell.







Every multimeter output is read through 100k (series) into a 74HCT165 input, with a
100k pull-down on the connector side:
- TTL-compatible thresholds for the 7404 outputs of the 70754 unit
- < 35 uA per line into the input clamp diodes when the isolated supply is off
  (USB unplugged), so the multimeter cannot back-power this board
- defined (all-zero) data when the board is not plugged into the multimeter

Which input reads which SKB pin was chosen for the shortest board layout
(scripts/optimise.py); the firmware uses the table below.

Serial stream at MISO after a LATCH pulse; bit 0 is shifted out first.
Numbers are SKB pins, "=1" / "=0" are hard-wired marker bits (chain length check).

bit 0.. 7	U10 D7..D0 :	2	18	35	36	34	1	3	19
bit 8..15	U11 D7..D0 :	22	38	5	20	21	4	6	23
bit 16..23	U12 D7..D0 :	26	27	24	25	7	8	9	10
bit 24..31	U13 D7..D0 :	14	31	29	28	12	11	13	30
bit 32..39	U14 D7..D0 :	=1	=0	=0	33	16	32	17	15

SKB 38 is the read-back of the FRONT PANEL LOCKOUT line (high = DVM powered).

Generated by scripts/gen_sch.py
Solartron 7075 USB interface (RP2354A)

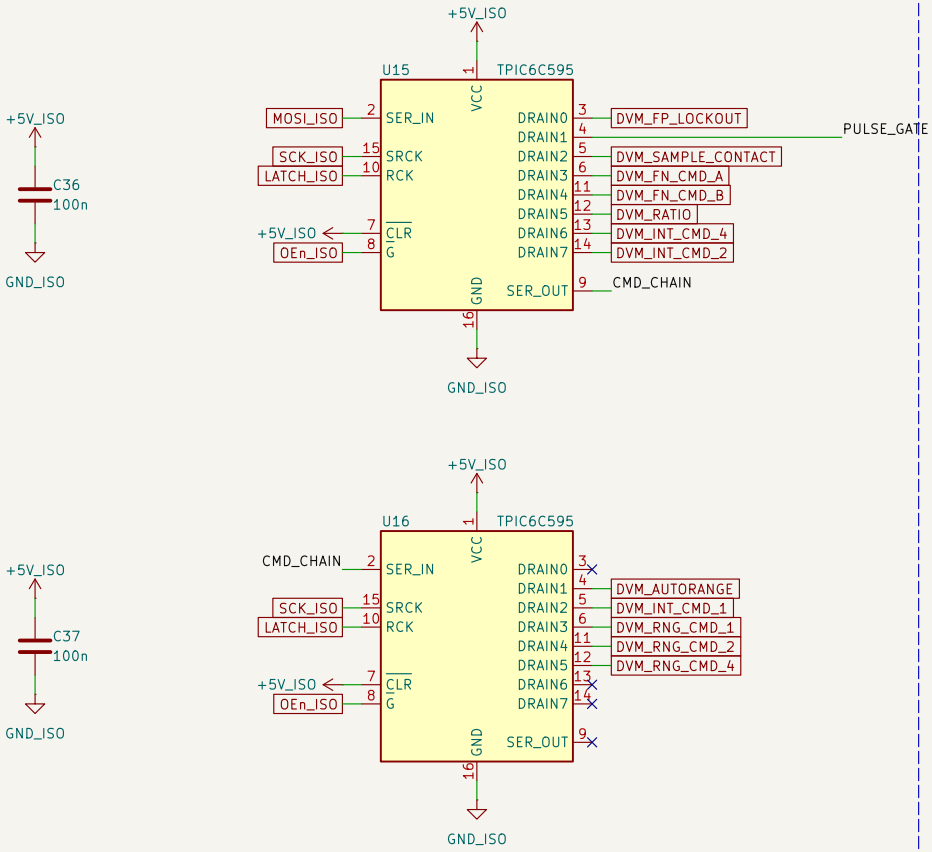
Sheet: /DVM_INPUTS/
File: dvm_inputs.kicad_sch

Title: Multimeter connector and status / BCD data inputs

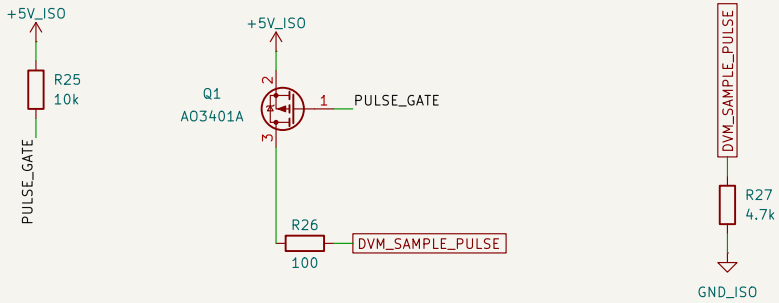
Size: A3	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 4/5

Open-drain command drivers

output off = line released = logic 1 at the multimeter



SAMPLE pulse driver (SKB pin 40: positive pulse > 100 us)



Command word = the last 16 bits clocked into MOSI before LATCH rises (MSB first):

bit 0 -> SKB 38 FRONT PANEL LOCKOUT	bit 8 -> spare
bit 1 -> SKB 40 SAMPLE pulse (via Q1)	bit 9 -> SKB 47 AUTORANGE
bit 2 -> SKB 39 SAMPLE (contact to 0 V)	bit 10 -> SKB 46 INTEGRATION TIME (1)
bit 3 -> SKB 42 FUNCTION	bit 11 -> SKB 48 RANGE (1)
bit 4 -> SKB 43 FUNCTION	bit 12 -> SKB 49 RANGE (2)
bit 5 -> SKB 41 RATIO	bit 13 -> SKB 50 RANGE (4)
bit 6 -> SKB 44 INTEGRATION TIME (4)	bit 14 -> spare
bit 7 -> SKB 45 INTEGRATION TIME (2)	bit 15 -> spare

A 1 turns the open-drain output ON, i.e. pulls the SKB line to logic 0 (SKB 40: a 1 drives the line HIGH through Q1). The multimeter has its own 4.7k pull-ups to +5 V on every command line.

_G (OEn_ISO) is pulled high until the MCU enables it, and all drains are off without isolated power, so an unpowered / unconfigured board leaves the multimeter exactly as if nothing were plugged in.

Generated by scripts/gen_sch.py

Solartron 7075 USB interface (RP2354A)

Sheet: /DVM_OUTPUTS/

File: dvm_outputs.kicad_sch

Title: Multimeter command outputs (open-drain)

Size: A4	Date: 2026-11-11	**Rev: A**
KiCad E.D.A. 10.0.8		Id: 5/5